Writing Systems in East and Southeast Asia

Francis Bond

based on Goddard Chapter 6

2026

Overview

1. Introduction to Writing Systems
2. Types of Writing Systems
3. Abjad: Jawi
4. Alphabet: Rumi
5. Abugida: Thai
6. Featural: Hangul
7. Logographic: Chinese
8. Japanese: A Multiscript System
9. Vietnamese
10. Unicode
11. Calligraphy

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Writing Makes Speech Visible

logographic	a symbol for a word or morpheme
phonographic	a symbol for a sound: phoneme or syllable

- Language is primarily spoken, but writing is stable and visible
- The history of a script is a history of contact

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Six Kinds of Script, All in This Region

type	a symbol writes ...	example
logographic	a morpheme	Chinese 山 *shān*
abjad	a consonant; vowels optional	Jawi (Arabic script)
abugida	a consonant with an inherent vowel; other vowels as signs around it	Thai, Khmer, Burmese
alphabet	consonants and vowels alike	Rumi, Vietnamese
syllabary	a syllable (strictly, a mora)	Japanese kana
featural	features, inside the letter shapes	Korean Hangul

Japanese **mixes** systems: kanji, two kana, and Latin letters

Three Great Source Traditions

Chinese c. 1200 BCE	Brahmi 3rd c. BCE	Arabic 14th c. CE, with Islam
Japanese kanji, kana	Khmer, Thai, Lao	Malay Jawi
Korean hanja	Mon, Burmese	
Vietnamese chữ Nôm	Javanese, Balinese	

- Brahmi reached Southeast Asia through the south Indian Pallava script
- Latin came later, with missionaries and colonial rule (Rumi, Vietnamese)
- One **invention**: Korean Hangul

Where Did Brahmi Come From?

- First securely dated under **Ashoka**, 3rd century BCE
- Its neighbour **Kharoṣṭhī** (right to left) comes from **Aramaic**, the Persian empire's administrative script
- Brahmi most likely had a Semitic model too, **reworked** by Indian scholars: left to right, vowel signs, long vs short vowels
- Descent from the Indus script is a minority view: that script is undeciphered and 1,500 years older

Devanagari	Thai	Lao	Khmer	Burmese	
क	ก	ກ	ក	က	all *ka*

Salomon (1998); Scharfe (2002)

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Jawi: Arabic Script for Malay

السلام عليكم *as-salām ’alaykum* “peace be upon you”
جاوي *Jawi* < *al-Jazā’ir al-Jāwī* “Java archipelago”

- An **abjad**: written right to left, short vowels usually not written
- *Jawi* also means “Malay”: *Bahasa Jawi* “Malay language”

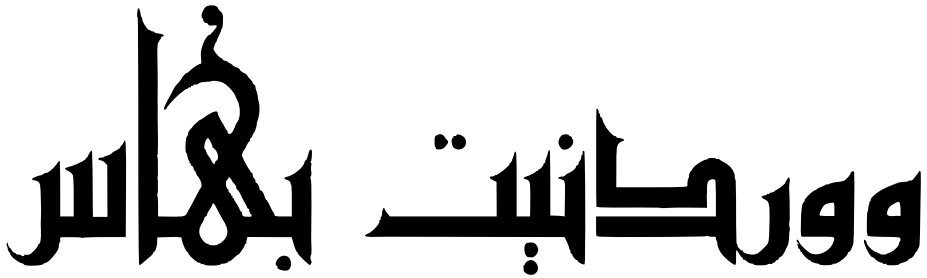
Six New Letters for Malay

for native Malay sounds			
چ	ca	/t͡ʃ/	
ڠ	nga	/ŋ/	
ڤ	pa	/p/	
ک	ga	/g/	
ڽ	nya	/ɲ/	*nyonya*
for loanwords			
ۏ	va	/v/	

Added to the Arabic abjad: 28 letters, plus hamza, tāʾ marbūṭah and lām-alif

Jawi Today

- Letters change shape by position in the word
- Written with a **reed pen** (qalam), giving the thick–thin strokes
- In decline, but still taught in Malaysian schools, and used in religion and decoration



Wordnet Bahasa

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Rumi: One Spelling for Two Countries

before 1972	after	
ketjap (Ind.), *kechap* (Mal.)	*kecap*	/tʃ/ is *c*
njonja (Ind.), *nyonya* (Mal.)	*nyonya*	
di-buat	*dibuat*	prefix joined
di-rumah	*di rumah*	preposition apart
ke-barat^{2}*-an*	*kebarat-baratan*	"westernized"

- Latin letters came with colonial rule
- Malaysian spelling followed English, Indonesian followed Dutch
- A common spelling since 1972

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Thai: An Abugida

คน	*khon*	person: no vowel written
สวัสดี	*sawàtdii*	hello

- Each consonant carries an **inherent vowel**; other vowels are signs above, below, before or after it
- No spaces between words
- From Old Khmer, which goes back to Brahmi via Pallava

Thai Tones: Letter Class Matters

high-class ข			low-class ค		
ขา	*khǎa*	leg	คา	*khaa*	embedded
ข่า	*khàa*	galangal	ค่า	*khâa*	cost
ข้า	*khâa*	slave	ค้า	*kháa*	trade
ขาด	*khàat*	lack	คาด	*khâat*	strap

- Same /kh/, same tone mark, different tone
- The tone depends on the consonant's class, the tone mark and the syllable
- Complicated but deterministic (Smalley 1994: 184)

Thai Vowels

อะ	อั	อา	อิ	อี	อึ	อื
a	a	a	i	i	ue	ue
[a]	[a]	[aː]	[i]	[iː]	[ɯ]	[ɯː]
อือ	อุ	อู	เอะ	เอ็	เอ	แอะ
ue	u	u	e	e	e	ae
[ɯː]	[u]	[uː]	[e]	[e]	[eː]	[ɛ]
แอ็	แอ	โอะ	โอ	เอาะ	อ็อ	ออ
ae	ae	o	o	o	o	o
[ɛ]	[ɛː]	[o]	[oː]	[ɔ]	[ɔ]	[ɔː]
อ็	เออะ	เออ	เอิ			
o	oe	oe	oe			
[ɔː]	[ɤʔ]	[ɤː]	[ɤː]			

Thai Diphthongs

เอียะ	เอีย	เอือะ	เอือ	อัวะ	อัว	อว
ia	ia	uea	uea	ua	ua	ua
[ia?]	[ia]	[ɯa?]	[ɯa]	[ua?]	[ua]	[ua]
อิว	เอ็ว	เอว	แอว	เอา	อาว	เอียว
io	eo	eo	aeo	ao	ao	iao
[iu/iw]	[eu/ew]	[e:u/e:w]	[ɛ:u/ɛ:w]	[au/aw]	[a:u]	[iau/iaw]
อัย	ไอ	ไอ	ไอย	อาย	อ็อย	ออย
ai	ai	ai	ai	ai	oi	oi
[ai/aj]	[ai/aj]	[ai/aj]	[ai/aj]	[a:i/a:j]	[ɔi/ɔj]	[ɔ:i/ɔ:j]
โอย	อุย	เอย	อวย	เอือย	อำ	ฤ
oi	ui	oei	uai	ueai	am	rue
[o:i/o:j]	[ui/uj]	[ɤ:i/ɤ:j]	[uai/uaj]	[ɯai/ɯaj]	[am]	[rɯ/ri]
ฤๅ	ฦ	ฦๅ	อๆ	อ์		
rue	lue	lue	m̩	silences		
[rɯ:]	[lɯ]	[lɯ:]	[~]	consonant		

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Korean Hangul



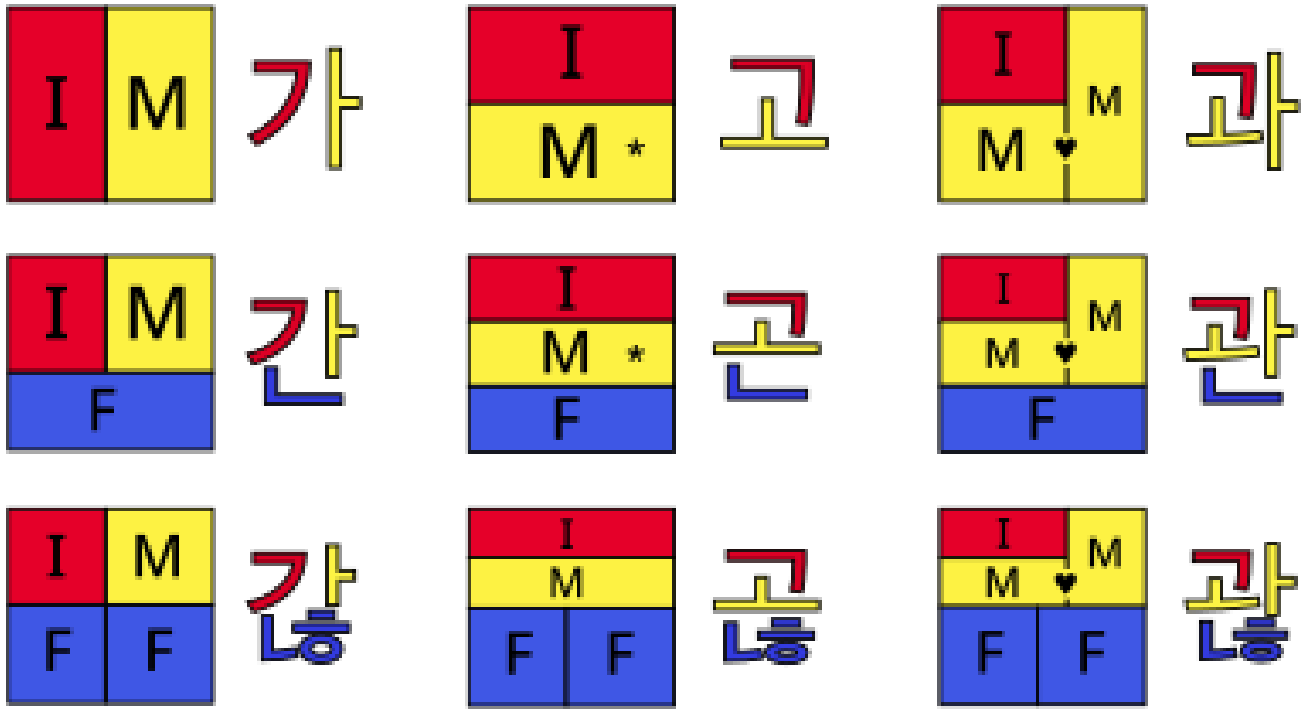
Image from Wikipedia (2024)

Hangul: Invented, Not Inherited

안녕하세요 *annyeonghaseyo* “hello”

- Invented by **King Sejong** in the 15th century, and documented in the *Hunmin jeongeum* “The correct sounds for the instruction of the people”
- Looked down on by scholars, who used **hanja**, as a script for women, children and the uneducated
- Not widely adopted until the 19th century

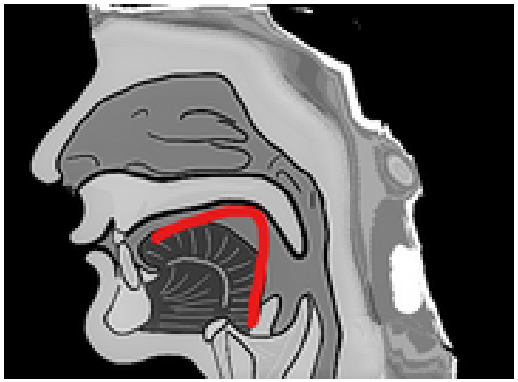
One Syllable, One Block



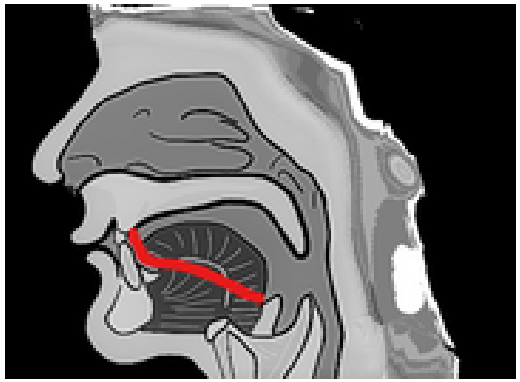
Vowels are vertical or horizontal strokes; consonant shapes reflect where they are articulated

Image from Wikipedia (2024), cropped

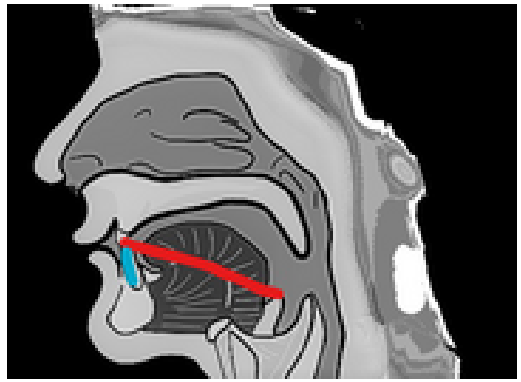
The Shape Shows the Articulation



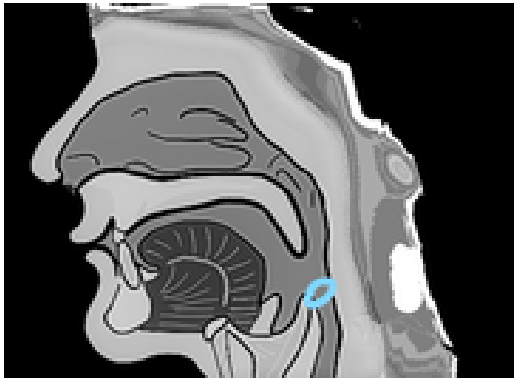
ㄱ /k/



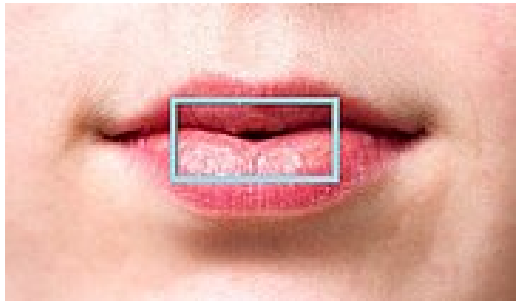
ㄴ /n/



ㅅ /s/



ㅇ /ŋ/



ㅁ /m/

Images from Wikipedia (2024)

Add a Stroke, Add a Feature

	basic	plain	aspirated	tense
velar		ㄱ k	ㅋ k^h	ㄲ k͈
coronal	ㄴ n	ㄷ t	ㅌ t^h	ㄸ t͈
bilabial	ㅁ m	ㅂ p	ㅍ p^h	ㅃ p͈
sibilant		ㅅ s		ㅆ s͈
palatal		ㅈ tɕ	ㅊ $tɕ^h$	ㅉ tɕ͈

ㅇ is silent at the start of a syllable and ŋ at the end; ㅎ is h

“Before the Morning Is Over”

A wise man can acquaint himself with them before the morning is over; even a stupid man can learn them in the space of ten days.

Hunmin Jeongeum Haerye, postface of Jeong Inji, p. 27a,
translated by Gari K. Ledyard, *The Korean Language Reform of 1446*, p. 258

- Well motivated, but not a direct map of phonetic features
- Sounds change in combination, so the spelling is morphophonemic

Cia-Cia: Hangul for an Austronesian Language

who	80,000 speakers on Buton Island, SE of Sulawesi
before	a Jawi-like script (Gundhul), now mainly Rumi
2007	a Hangul orthography by Chun Tai-hyun (HUFS, Seoul)
2009	taught, from a textbook by Lee Ho-young (SNU)
2012	the Korean institute leaves; teaching dwindles
2021	a Cia-Cia dictionary in Hangul
2022	the institute returns

- For /v/ it revives the obsolete jamo ㅱ *kapyeounpieup* “light pieup”
- Cia-Cia (바하사찌아찌아) is Celebic, Malayo-Polynesian

Cia-Cia in Use



From Language of the Month November 2024 – Cia-Cia by L S, The National Museum of Language (online virtual museum)

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Chinese: 3,000 Years and Counting

山	*shān*	mountain
你好	*nǐ hǎo*	hello (“you good”)

- Grew out of pictograms; in continuous use for over 3,000 years
- Characters mostly write **morphemes**
- A force for cultural and political unity, adopted by Japanese, Korean and Vietnamese
- Simplified in the People’s Republic in the 1950s

The Character for "Horse" 马

Type of Characters	Descriptions
金文 	Bronze script Jin wen 15th - 11th centuries B.C.E.
甲骨文 	Oracle-bone script Jia gu wen 12th - 11th centuries B.C.E.
大篆 	Large-seal script da chuan c. 8th century B.C.E.
小篆 	Small-seal script xiao chuan 2nd century B.C.E.
隸書 	Clerical script li shu 2nd century C.E.
楷書 	Standard script k'ai shu since c. 4th century C.E.
行書 	Running script Xing shu since c. 4th century C.E.
草書 	Cursive script Chao shu since c. 4th century C.E.

Logographic Writing: Chinese

Pictographs

女	子	口	日	月	山	川	豕	目	心	雨	田	木	龜
woman	child	mouth	sun	moon	mountain	river	pig	eye	heart	rain	field	tree	turtle

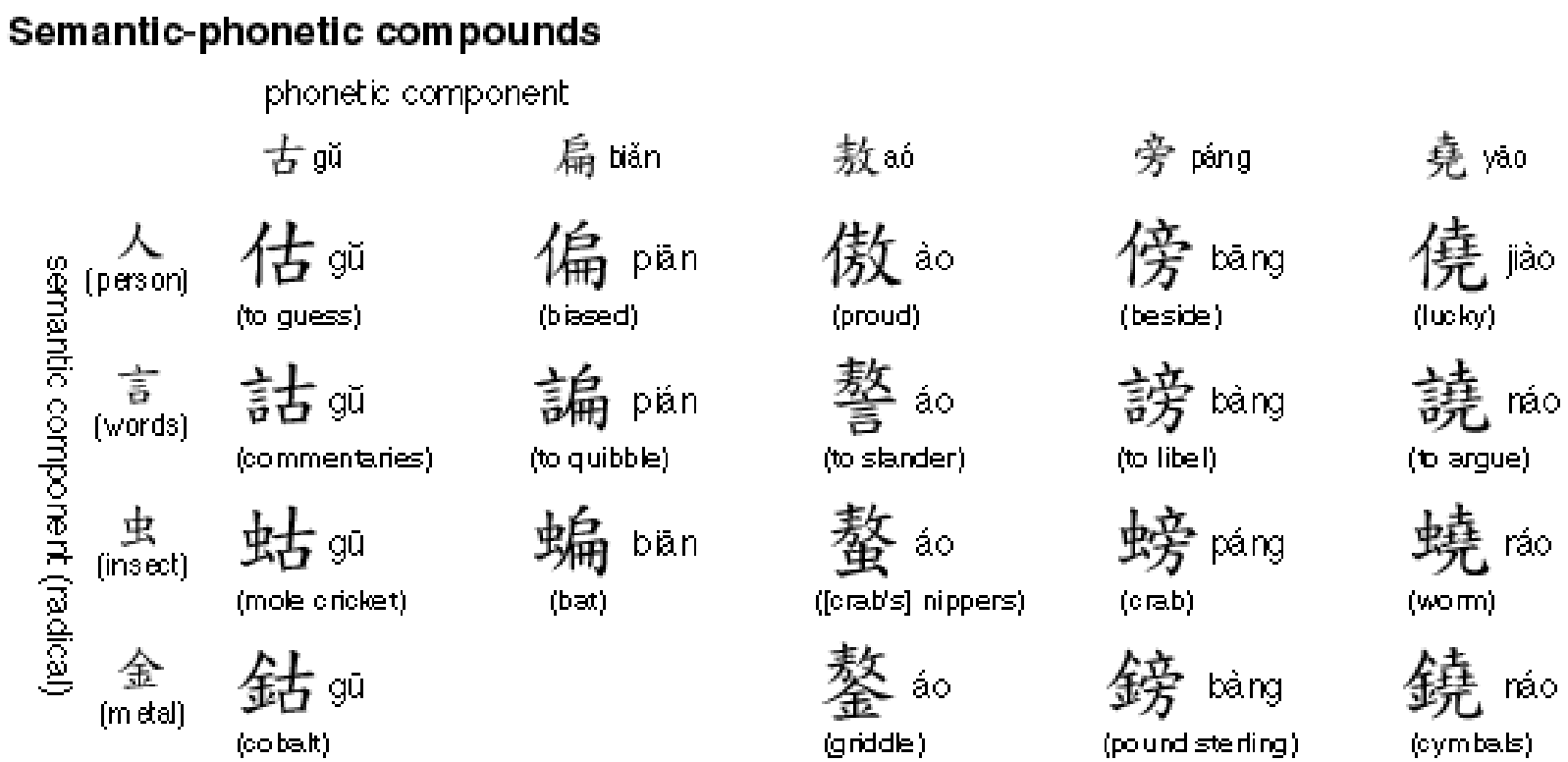
Ideographs

一	二	三	上	下	中	力	凸	凹
one	two	three	above	below	middle	stength (plough)	convex	concave

Compound Pictographs / Compound Ideographs

好	安	明	家	思	牢	雷	男
good (woman + child)	peaceful (woman under a roof)	bright (sun + moon)	home/family (pig under a roof)	thought (heart + field)	prison (cow under a roof)	thunder (rain cloud over a field)	man/male (field + strength)

Most Characters Are Semantic + Phonetic



- The **radical** hints at meaning, the **phonetic** at sound
- Both hints are irregular, so characters must be memorized
- 97% of characters are phonetic compounds (Sproat 2010); other estimates are lower (Wang, p.c. 2013)

Chữ Nôm: The Same Trick for Vietnamese

巴	*ba*	sound
三	“three”	meaning
= □	*ba*	“three”

The finished character is in CJK Extension B, which these fonts lack, so you get a **tofu** box

Chinese Characters for Other Sinitic Languages

- The script began with **Old Chinese**, c. 1200 BCE
- The modern standard written language is based on **Mandarin**, so phonetic hints fit Mandarin, not Cantonese or Hokkien
- Literacy in Cantonese or Taiwanese assumes the Mandarin standard
- Many Cantonese words and particles have no standard character

Writing Cantonese: Three Strategies

new characters	哋 *dei6*	plural: 口 “mouth” + 地 *dei6*
sound-alikes	果度 for 嗰啲	“there”
Latin letters	D for 啲	“of”

Because Cantonese is mainly spoken, its spelling varies a lot

Workarounds in Practice

你	喺	嗰啲	好	喇,
你	o係	果度	好	la,
you	be.at	there	good	SFP

千祈	咪	搞	佢	啲	嘢
千祈	咪	搞	佢	D	野
on.no.account	don't	mess.with	3SG	POSS	stuff

"You'd better stay there, and under no circumstances mess with their stuff." [yue]

Second line: o for the 口 radical, D for 啲, 果度 for 嗰啲

Cantonese Borrows More

Cantonese		English	Mandarin	
BB	*bi4 bi1*	baby	婴儿	*yīng'ér*
巴士	*baa1 si2*	bus	公交车	*gōngjiāochē*
拜拜	*baai1 baai3*	bye bye	再见	*zàijiàn*
沙展	*saa1 zin2*	sergeant	警长	*jǐngzhǎng*

English influence: Cantonese uses more loans (as does Singapore Mandarin), alongside Mandarin vocabulary

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Japanese Mixes Scripts

kanji	lexical roots	山 *yama* “mountain”
hiragana	endings, particles	こんにちは *konnichiwa*
katakana	loans, mimetics, emphasis	シャツ *shatsu* “shirt”
rōmaji	abbreviations, brands	T

(1) 彼女 は T シャツ を 着て いる
kanojo wa T-shatsu o ki-te iru
she TOP T-shirt ACC wear-CONJ PROG
“She is wearing a T-shirt.” [jpn]

Two Kana, Two Origins

hiragana	cursive forms of whole kanji
katakana	parts of kanji

- Both chose kanji with a suitable pronunciation
- Both are **moraic syllabaries**: ん and small つ each count as one mora

Hiragana: A Moraic Syllabary

平仮名（ひらがな） hiragana

a	あ	安	i	い	以	u	う	宇	e	え	衣	o	お	於
ka	か	加	ki	き	幾	ku	く	久	ke	け	計	ko	こ	己
sa	さ	左	shi	し	之	su	す	寸	se	せ	世	so	そ	曽
ta	た	太	chi	ち	知	tsu	つ	川	te	て	天	to	と	止
na	な	奈	ni	に	仁	nu	ぬ	奴	ne	ね	祢	no	の	乃
ha	は	波	hi	ひ	比	fu	ふ	不	he	へ	部	ho	ほ	保
ma	ま	末	mi	み	美	mu	む	武	me	め	女	mo	も	毛
ya	や	也				yu	ゆ	由				yo	よ	与
ra	ら	良	ri	り	利	ru	る	留	re	れ	礼	ro	ろ	呂
wa	わ	和	wi	ゐ	為				we	ゑ	恵	wo	を	遠
												n	ん	无

Katakana: A Moraic Syllabary

片仮名（カタカナ） katakana

a	ア	阿	i	イ	伊	u	ウ	宇	e	エ	江	o	オ	於
ka	カ	加	ki	キ	幾	ku	ク	久	ke	ケ	介	ko	コ	己
sa	サ	散	shi	シ	之	su	ス	須	se	セ	世	so	ソ	曽
ta	タ	多	chi	チ	千	tsu	ツ	川	te	テ	天	to	ト	止
na	ナ	奈	ni	ニ	二	nu	ヌ	奴	ne	ネ	祢	no	ノ	乃
ha	ハ	八	hi	ヒ	比	fu	フ	不	he	ヘ	部	ho	ホ	保
ma	マ	万	mi	ミ	ミ	mu	ム	牟	me	メ	女	mo	モ	毛
ya	ヤ	也				yu	ユ	由				yo	ヨ	與
ra	ラ	良	ri	リ	利	ru	ル	流	re	レ	礼	ro	ロ	呂
wa	ワ	和	wi	ヰ	井				we	ヱ	恵	wo	ヲ	乎
												n	ン	无

Extending the Kana

dakuten 濁点 ゛	voicing	か ka → が ga
handakuten 半濁点 ゜	voiceless /p/	は ha → ぱ pa
yōon 拗音	palatalized	き + や → きゃ kya
gōyōon 合拗音	labialized	くわ kwa (Okinawan)
sokuon 促音	geminate	さっか *sakka*

濁点 *dakuten* is the mark; 濁音 *dakuon* is the voiced sound it writes

Kanji Tell Homophones Apart

こうしょう *kōshō*	
交渉	negotiation
工廠	arsenal, workshop
鉱床	ore deposit

- Few distinct syllables, so many homophones: WWWJDIC lists 44 entries for *kōshō*
- Context picks the word, and the kanji

One Character, Many Readings: 明

reading	from		example
kun	native Japanese	*aka(rui)*	明るい bright
go-on	Wu, 5th–6th c.	*myō*	明日 *myōnichi* tomorrow
kan-on	Tang, 7th–9th c.	*mei*	説明 *setsumei* explanation
tō-on	Song and Ming	*min*	明朝 *minchō* Ming dynasty
nanori	names	*Akira*	

Kanjidic gives 明 3 on, 10 kun and 11 name readings; Chinese and Korean have one each: *míng*, *myeong*

`https://www.edrdg.org/cgi-bin/wwwjdic/wwwjdic?1B`

Why So Many Readings?

- Kanji arrived from the 5th century, **with** Chinese pronunciations: **on** readings, in waves from different periods and regions
- Many already had a native word: **kun** readings
- Compounds tend to use on (火山 *ka-zan* “volcano”), single kanji kun (山 *yama*)

The Cost: Memory, and Reforms

1946	**Tōyō** kanji: 1,850
1981	**Jōyō** kanji: 1,945
2010	**Jōyō** kanji, revised: 2,136

- An educated adult knows about 2,000 kanji, each with several readings: years of study
- Names and literature still use rarer kanji
- Typing means choosing among homophones; handwriting skills suffer

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Vietnamese: Three Scripts in Two Thousand Years

111 BCE – 20th c.	Classical Chinese (*văn ngôn*) in *chữ Hán*
13th – 20th c.	*chữ Nôm*: Chinese-style characters for Vietnamese
17th c. on	a Latin alphabet, from Portuguese missionaries
1910	the alphabet compulsory in schools (French rule)

At first the alphabet was used mainly in the Catholic community

The Vietnamese Alphabet

7 letters with diacritics	‹ă› ‹â› ‹ê› ‹ô› ‹ơ› ‹ư› ‹đ›
5 tone marks	‹à› ‹á› ‹ả› ‹ã› ‹ạ›; level tone unmarked
and they stack	*nhất* “first”

- 29 letters; sound change means different letters, digraphs and trigraphs now spell the same sound
- The North has 6 tones; the South has merged *hỏi* ‹ả› and *ngã* ‹ã› **in speech** (the spelling still distinguishes them)

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Unicode: Code Points Are Not Glyphs

- Unicode gives each character a **code point**; the **font** draws it
- **Han unification**: Chinese, Japanese and Korean share code points, though local shapes differ



all U+9AA8

Simplified Traditional Japanese Korean

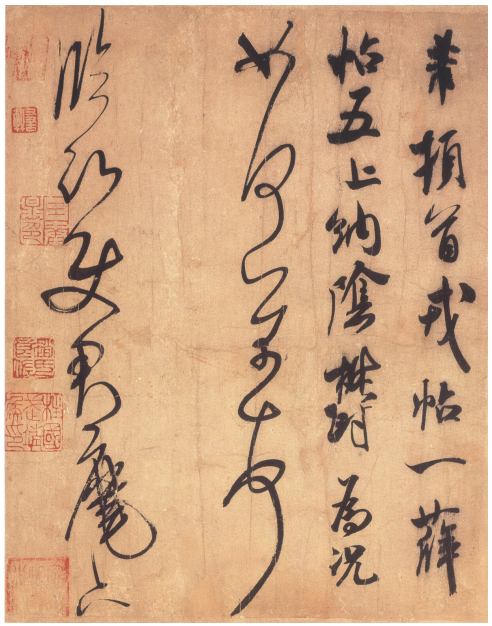
- Rare characters need a font that has them: obsolete  for Cia-Cia, and chữ Nôm (Extension B), which our fonts lack: □

Outline

1. Introduction to Writing Systems
2. Types of Writing Systems
3. Abjad: Jawi
4. Alphabet: Rumi
5. Abugida: Thai
6. Featural: Hangul
7. Logographic: Chinese
8. Japanese: A Multiscript System
9. Vietnamese
10. Unicode
11. Calligraphy

Calligraphy in East Asia

Valued above painting; styles run from seal and clerical script to standard, running and grass



Mi Fu (1051–1107)



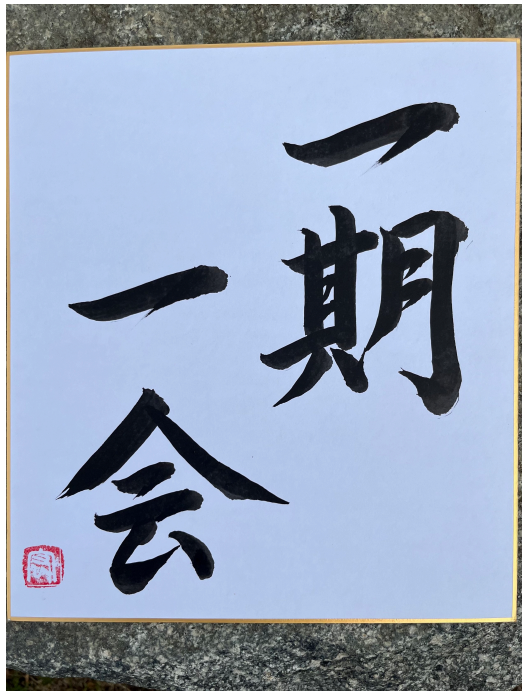
Emperor Huizong of Song (1082–1135)

Japanese Calligraphy

Grass style and Zen calligraphy: fluid and abstract



別無工夫 *betsu-naku kuufuu*
nothing else but focus



一期一会 *ichi-go ichi-e*
one lifetime, one chance

Jawi Calligraphy

Based on Arabic calligraphy, and often very ornate



Arabic Calligraphy in Diwani Style



يَا أَيُّهَا النَّبِيُّ إِنَّا أَرْسَلْنَاكَ شَاهِدًا وَمُبَشِّرًا وَنَذِيرًا وَدَاعِيًا إِلَى اللَّهِ

"O Prophet, indeed We have sent you as a witness and a bringer of good tidings and a warner, and as one who invites to Allah."

Mehmet Izzet al-Karkuki (1841–1904)

Diwani was developed under the early Ottoman Turks (16th – early 17th century)

Summary

- The region has **every type** of script: logographic, abjad, abugida, alphabet, syllabary, featural
- Most descend from **three traditions**: Chinese, Brahmi, Arabic — plus Latin, and one invention
- Scripts are **hard to change**: most countries have dropped or limited Chinese characters, but reprinting everything is a big job

Acknowledgements and References

Images from Omniglot (`https://omniglot.com/`) unless otherwise stated

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Hartmut Scharfe. 2002. Kharoṣṭī and Brāhmī. *Journal of the American Oriental Society*, 122(2):391–393.